

# Maharashtra Public Service Commission Assessment

|                                   |                                                                                        |
|-----------------------------------|----------------------------------------------------------------------------------------|
| <b>Subject Name :</b>             | Assistant Professor Chemistry Group A and Lecturer Chemistry Group B collegiate branch |
| <b>Duration :</b>                 | 60                                                                                     |
| <b>Number of Questions :</b>      | 100                                                                                    |
| <b>Total Marks :</b>              | 200                                                                                    |
| <b>Display Marks:</b>             | Yes                                                                                    |
| <b>Maximum Instruction Time :</b> | 0                                                                                      |
| <b>Is Section Default? :</b>      | No                                                                                     |

Question Number : 1 Question Id : 4646621301 Question Type : MCQ

In which period of Periodic table are there the fewest elements present ?

Options :

1. Seventh Period
2. Second Period
3. First Period
4. Sixth Period

Question Number : 2 Question Id : 4646621302 Question Type : MCQ

Among the following acid-base theory which theory related with only proton ?

Options :

1. Arrhenius theory
2. Lowry-Bronsted theory
3. Lewis theory
4. Lux-Flood theory

Question Number : 3 Question Id : 4646621303 Question Type : MCQ

Which of the following is a soft acid ?

$\text{Be}^{2+}$ ,  $\text{Mg}^{2+}$ ,  $\text{Ca}^{2+}$ ,  $\text{Cd}^{2+}$

Options :

1.  $\text{Be}^{2+}$
2.  $\text{Mg}^{2+}$
3.  $\text{Ca}^{2+}$
4.  $\text{Cd}^{2+}$

Question Number : 4 Question Id : 4646621304 Question Type : MCQ

The concentration of  $\text{H}^+$  and  $\text{OH}^-$  ions in aqueous solutions are frequently very small number which is practically measured by :

Options :

1. pH - meter
2. Potentiometer
3. Conductivity meter
4. Spectrophotometer

Question Number : 5 Question Id : 4646621305 Question Type : MCQ

The **correct** order of second ionization enthalpies is :

Options :

1.  $\text{B} > \text{C} > \text{N} > \text{Be}$
2.  $\text{N} > \text{C} > \text{Be} > \text{B}$
3.  $\text{B} > \text{N} > \text{C} > \text{Be}$
4.  $\text{N} > \text{B} > \text{C} > \text{Be}$

Question Number : 6 Question Id : 4646621306 Question Type : MCQ

Generally on moving across the period, the basic character of oxides gradually changes first into amphoteric and finally in acidic, which of the following statement is **correct** ?

- (a)  $\text{P}_2\text{O}_5$  is acidic in nature
- (b)  $\text{SiO}_2$  is strongly basic in nature
- (c)  $\text{Na}_2\text{O}$  is strongly basic in nature
- (d)  $\text{Al}_2\text{O}_3$  is basic in nature

Options :

- 1. Only (a)
- 2. Only (b)
- 3. Only (c)
- 4. (a) and (c)

Question Number : 7 Question Id : 4646621307 Question Type : MCQ

HSAB principle is **not** useful for explaining :

Options :

1. Stability of compounds having same ligands
2. Nature of bonding in complex compounds by ambidentate ligands
3. Bond order
4. Stability of compounds having different ligands

Question Number : 8 Question Id : 4646621308 Question Type : MCQ

Carbon atoms have the ability to bond to themselves to a greater extent than those of any other element known as :

Options :

1. Catenation
2. Carbocation
3. Intercalation
4. Carbanion

Question Number : 9 Question Id : 4646621309 Question Type : MCQ

Wood charcoal is used in gas masks because :

Options :

1. It is poisonous
2. It liquifies gas
3. It is porous
4. It adsorbs gases

Question Number : 10 Question Id : 4646621310 Question Type : MCQ

According to the principle of the \_\_\_\_\_ spectroscopy certain atomic nuclei have properties of spin and magnetic moment. Exposure of these nuclei to magnetic field leads to splitting of their energy level.

Options :

1. NMR
2. IR
3. UV
4. Visible

Question Number : 11 Question Id : 4646621311 Question Type : MCQ

Misch metals are :

Options :

1. Alloys of rare earth metals
2. Alloys of d-block elements
3. Special steels
4. Alloy of Al, Mg and Gd

Question Number : 12 Question Id : 4646621312 Question Type : MCQ

Which one of the following divalent ions is expected to have the minimum octahedral ionic radius ?

Options :

1.  $\text{Ti}^{2+}$
2.  $\text{Cr}^{2+}$
3.  $\text{V}^{2+}$
4.  $\text{Mn}^{2+}$

Question Number : 13 Question Id : 4646621313 Question Type : MCQ

The complex ion developing the highest crystal field splitting energy is :

Options :

1.  $[\text{Co}(\text{NH}_3)_6]^{2+}$
2.  $[\text{Rh}(\text{NH}_3)_6]^{3+}$
3.  $[\text{Ir}(\text{NH}_3)_6]^{3+}$
4.  $[\text{Co}(\text{NH}_3)_6]^{3+}$

Question Number : 14 Question Id : 4646621314 Question Type : MCQ

The **correct** option for oxidation state(s) of Nb in the cluster  $\text{Na}_4[\text{Nb}_6\text{Cl}_{18}]$  is :

Options :

1. Two are in  $3+$  state and four are in  $2+$  state
2. All are in  $2+$  state
3. All are in  $3+$  state
4. Three are in  $2+$  state and three are in  $3+$  state



Question Number : 15 Question Id : 4646621315 Question Type : MCQ

The CFSE for a high-spin octahedral complex of a  $d^6$  ion is :

Options :

1.  $-0.6 \Delta_o$
2.  $1.2 \Delta_o$
3.  $1.8 \Delta_o$
4.  $-0.4 \Delta_o$

Question Number : 16 Question Id : 4646621316 Question Type : MCQ

What is the oxidation state of phosphorus in the oxoacid  $H_3PO_4$ , and what is the geometry around the phosphorus atom ?

Options :

1. +3, Tetrahedral
2. +5, Tetrahedral
3. +3, Trigonal pyramidal
4. +5, Trigonal planar

Question Number : 17 Question Id : 4646621317 Question Type : MCQ

The Wilkinson's catalyst is :

Options :

1.  $[\text{RhCl}(\text{PPh}_3)_3]$
2.  $[\text{Rh}(\text{PPh}_3)_3\text{Cl}]$
3.  $[\text{RhCl}(\text{PEt}_3)_3]$
4. None of the above

Question Number : 18 Question Id : 4646621318 Question Type : MCQ

In Ziegler-Natta polymerization, the catalyst system typically include :

Options :

1.  $\text{Ni}(\text{CO})_4$  and water
2.  $\text{AlCl}_3$  and benzene
3. Pt and  $\text{H}_2$
4.  $\text{TiCl}_4$  and an organoaluminium compound

Question Number : 19 Question Id : 4646621319 Question Type : MCQ

In Mossbauer spectroscopy, the main hyperfine interactions is/ are :

Options :

1. Isomer shift
2. Quadrupole splitting
3. Zeeman splitting
4. All of the above

Question Number : 20 Question Id : 4646621320 Question Type : MCQ

The fingerprint region in IR spectra ranges from :

Options :

1.  $1500 - 700 \text{ cm}^{-1}$
2.  $1950 - 1550 \text{ cm}^{-1}$
3.  $2260 - 2190 \text{ cm}^{-1}$
4. None of the above

Question Number : 21 Question Id : 4646621321 Question Type : MCQ

The iron in haemoglobin is in the :

Options :

1. III state
2. II state
3. IV state
4. I state

Question Number : 22 Question Id : 4646621322 Question Type : MCQ

During measurement of potential, the potential should be \_\_\_\_\_ for one of the electrode, such an electrode is termed as the reference electrode.

Options :

1. Constant
2. Variable
3. Low
4. High

Question Number : 23 Question Id : 4646621323 Question Type : MCQ

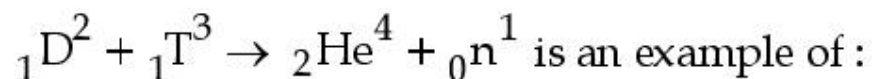
The common locating agent used to identify amino acids is :

Options :

1. Iodine vapour
2. Ninhydrin
3. Permanganate
4. Hydrogen sulphide

Question Number : 24 Question Id : 4646621324 Question Type : MCQ

The reaction –



Options :

1. Nuclear fission
2. Nuclear fusion
3. Artificial radioactivity
4. Radio disintegration

Question Number : 25 Question Id : 4646621325 Question Type : MCQ

In an applied magnetic field, the number of orientations of a nucleus with a spin number  $I$ , is given by the formula :

Options :

1.  $2nI + 1$
2.  $I + 2$
3.  $2I + 1$
4.  $2(I + 1)$

Question Number : 26 Question Id : 4646621326 Question Type : MCQ

The lifetime of a mercury drop of DME during its working should be :

Options :

1. 1 to 15 seconds
2. Less than 3 seconds
3. 3 to 6 seconds
4. More than 20 seconds

Question Number : 27 Question Id : 4646621327 Question Type : MCQ

In TGA compact sample will decompose at a \_\_\_\_\_ temperature than a loose sample.

Options :

1. Lower
2. Higher
3. Constant
4. None of the above

Question Number : 28 Question Id : 4646621328 Question Type : MCQ

Match the following :

- |                           |                            |
|---------------------------|----------------------------|
| (a) $sp$ hybridisation    | (i) Tetrahedral            |
| (b) $sp^2$ hybridisation  | (ii) Trigonal planer       |
| (c) $sp^3$ hybridisation  | (iii) Trigonal bipyramidal |
| (d) $sp^3d$ hybridisation | (iv) Linear                |

Options :

1. (a)-(iv), (b)-(i), (c)-(iii), (d)-(ii)
2. (a)-(iv), (b)-(ii), (c)-(i), (d)-(iii)
3. (a)-(iv), (b)-(iii), (c)-(i), (d)-(ii)
4. (a)-(iv), (b)-(i), (c)-(ii), (d)-(iii)



Question Number : 29 Question Id : 4646621329 Question Type : MCQ

Plateaus in a thermogram indicate :

Options :

1. Loss in weight
2. Heat of reaction
3. Thermal stability
4. Loss of water molecules

Question Number : 30 Question Id : 4646621330 Question Type : MCQ

If  $m$  and  $t$  are respectively the flow rate and drop time of mercury of the DME, the limiting diffusion current  $I_d$  is proportional to :

Options :

1.  $m^{2/3} t^{1/6}$
2.  $m^{1/6} t^{2/3}$
3.  $m^{3/2} t^{1/6}$
4.  $m^{1/6} t^{1/6}$

Question Number : 31 Question Id : 4646621331 Question Type : MCQ

Main group element show characteristic of both metals and non-metals known as :

Options :

1. Metal
2. Non-metal
3. Metalloids
4. None of the above

Question Number : 32 Question Id : 4646621332 Question Type : MCQ

The magnetic behaviour of transition metal complexes is studied using :

Options :

1. Conductometry measurement
2. Magnetic susceptibility measurement
3. XRD measurement
4. Warner theory measurement

Question Number : 33 Question Id : 4646621333 Question Type : MCQ

Nuclear Fission involves :

Options :

1. Combination of nuclei
2. Splitting of heavy nuclei
3. Formation of isotopes
4. Emission of electrons

Question Number : 34 Question Id : 4646621334 Question Type : MCQ

The condition for constructive interference of X-ray is :

Options :

1. Path difference =  $\lambda/2$
2. Path difference =  $n\lambda$
3. Path difference = 0 only
4. Path difference =  $2\lambda$  only

Question Number : 35 Question Id : 4646621335 Question Type : MCQ

The ratio  $d_{100} : d_{110} : d_{111} = 1 : \frac{1}{\sqrt{2}} : \frac{1}{\sqrt{3}}$ , then the crystal belongs to \_\_\_\_\_ lattice.

Options :

1. simple cubic
2. face centred cubic
3. body centred cubic
4. coplanar cubic

Question Number : 36 Question Id : 4646621336 Question Type : MCQ

Molybdenum forms body-centred cubic crystals whose density is  $10.3 \text{ gm cm}^{-3}$ . Calculate distance between the (110) planes.

Molar Mass of  $M_0 = 95.94 \text{ g mol}^{-1}$

Options :

1. 313.9 pm
2. 222.0 pm
3. 181.2 pm
4. 111.2 pm

Question Number : 37 Question Id : 4646621337 Question Type : MCQ

The EMF of an electrochemical cell becomes zero when the cell :

Options :

1. is heated strongly
2. reaches equilibrium
3. contains concentrated acid
4. has high conductivity

Question Number : 38 Question Id : 4646621338 Question Type : MCQ

The equilibrium constant for a cell with  $E^0 = 0.591\text{V}$ ;  $n = 2$  will be :

Options :

1.  $10^{10}$
2.  $10^{20}$
3.  $10^{15}$
4. 20

Question Number : 39 Question Id : 4646621339 Question Type : MCQ

$\Lambda_m = \Lambda_m^\circ - (A + B \Lambda_m^\circ) \sqrt{C}$  The Debye Huckel constant in the equation, which gives a measure of Asymmetric effect is \_\_\_\_\_.

Options :

1.  $\frac{8.10 \times 10^5}{(DT)^{1/2} \eta}$

2.  $\frac{8.40 \times 10^5}{(DT)^{3/2}}$

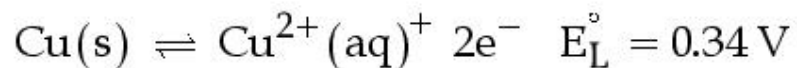
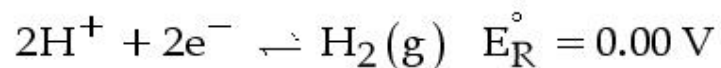
3.  $\frac{8.30 \times 10^5}{(DT)^{2/3}}$

4.  $\frac{8.20 \times 10^5}{(DT)^{3/2}}$

Question Number : 40 Question Id : 4646621340 Question Type : MCQ

Calculate the EMF of the following electrochemical cell at 25 °C :

$\text{Cu}, \text{Cu}^{2+} (a = 0.1\text{M}) \parallel \text{H}^+ (a = 0.01\text{M}), \text{H}_2 : \text{Pt} (0.95 \text{ atm})$



Options :

1.  $-0.520 \text{ V}$
2.  $-0.340 \text{ V}$
3.  $-0.428 \text{ V}$
4.  $-0.389 \text{ V}$

Question Number : 41 Question Id : 4646621341 Question Type : MCQ

The bond length decreases as bond order \_\_\_\_\_.

Options :

1. increases
2. decreases
3. zero
4. always negative





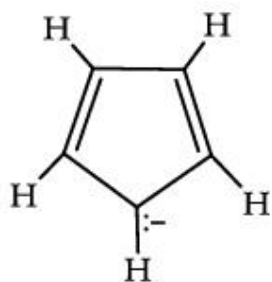
Question Number : 42 Question Id : 4646621342 Question Type : MCQ

Determine which one of the following molecule is non-aromatic, by using Huckel rule ?

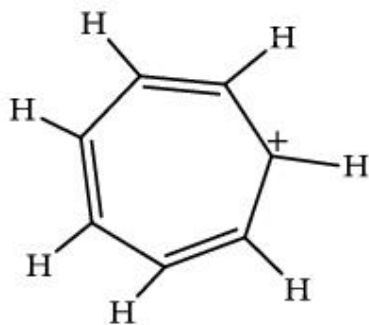
Options :



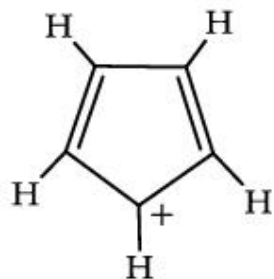
1.



2.



3.



4.

Question Number : 43 Question Id : 4646621343 Question Type : MCQ

Which Molecular orbital is formed by the head-on overlap of two atomic orbitals ?

Options :

1.  $\pi$  orbital
2.  $\delta$  orbital
3.  $\sigma$  orbital
4. Anti bonding orbital

**Question Number : 44 Question Id : 4646621344 Question Type : MCQ**

In addition polymerisation, polymer chain grows by :

**Options :**

1. Elimination of small molecule
2. Oxidation reactions only
3. Successive addition of monomer units
4. Crystallisation of monomers

**Question Number : 45 Question Id : 4646621345 Question Type : MCQ**

The median of the observations 5, 8, 10, 12, 14, 18, is :

**Options :**

1. 10
2. 11
3. 12
4. 13

Question Number : 46 Question Id : 4646621346 Question Type : MCQ

Standard deviation is a measure of :

Options :

1. Accuracy only
2. Precision or spread of data
3. Molecular mass
4. Reaction rate

Question Number : 47 Question Id : 4646621347 Question Type : MCQ

A chemical contains microscopic nitrogen bubbles uniformly dispersed in water. According to colloid classification, it would be :

Options :

1. Sol
2. Emulsion
3. Aerosol
4. Foam

Question Number : 48 Question Id : 4646621348 Question Type : MCQ

Colloid particles generally possess \_\_\_\_\_.

Options :

1. Uniform molecular size
2. Zero surface area
3. No Kinetic motion
4. Intermediate size between true solution and suspensions

Question Number : 49 Question Id : 4646621349 Question Type : MCQ

Which equation relates rate constant with ionic strength ?

Options :

1. Henderson equation
2. Nernst equation
3. Bronsted-Bjerrum equation
4. Clausius equation

Question Number : 50 Question Id : 4646621350 Question Type : MCQ

The unimolecular gas reactions could be understood only on the basis of the radiation hypothesis according to \_\_\_\_\_.

Options :

1. Van't Hoff
2. Perrin
3. Deniels and Johnsten
4. Trautz and Winkler

Question Number : 51 Question Id : 4646621351 Question Type : MCQ

The primary requirement for a photochemical reaction is \_\_\_\_\_.

Options :

1. High pressure
2. Absorption of radiation
3. High concentration
4. Catalyst

Question Number : 52 Question Id : 4646621352 Question Type : MCQ

Conductivity power of all ions produced by one more of an electrolyte in a given solution is \_\_\_\_\_.

Options :

1. Specific conductance
2. Equivalent conductance
3. Molar conductance
4. Ionic conductance

Question Number : 53 Question Id : 4646621353 Question Type : MCQ

Steady state approximation is valid when \_\_\_\_\_.

Options :

1. intermediate is highly stable and having long lifetime
2. intermediate concentration is higher than the reactant concentration
3. intermediate is highly reactive, having short lifetime and very low concentration
4. reaction occurs in single step

Question Number : 54 Question Id : 4646621354 Question Type : MCQ

The Boltzmann constant  $k$  is related to the Gas constant  $R$  by :

Options :

1.  $k = R N_A$

2.  $k = \cancel{R} / \cancel{N_A}$

3.  $k = N_A / \cancel{R}$

4.  $k = RT$

Question Number : 55 Question Id : 4646621355 Question Type : MCQ

Boltzmann distribution describes the distribution of molecules amongst :

Options :

1. different energy states

2. different phases

3. different isotopes

4. different reaction mechanisms



**Question Number : 56 Question Id : 4646621356 Question Type : MCQ**

During an Isothermal expansion of an ideal gas which thermodynamic quantity remains constant ?

**Options :**

1. Pressure
2. Volume
3. Temperature
4. Entropy

**Question Number : 57 Question Id : 4646621357 Question Type : MCQ**

The second law of thermodynamics predicts that the entropy of an isolated system :

**Options :**

1. Always decreases
2. Remains zero
3. Increases for spontaneous processes
4. Depends only on pressure

Question Number : 58 Question Id : 4646621358 Question Type : MCQ

Which of the following is **not** a state function ?

Options :

1. Pressure
2. Temperature
3. Internal energy
4. Work

Question Number : 59 Question Id : 4646621359 Question Type : MCQ

Identify an extensive property amongst the following :

Options :

1. Viscosity
2. Heat capacity
3. Density
4. Surface tension

Question Number : 60 Question Id : 4646621360 Question Type : MCQ

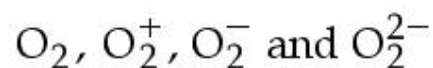
According to Huckel Molecular Orbital theory, Benzene is specially stable because it contains :

Options :

1. Localised  $\sigma$  electrons only
2. Six delocalised  $\pi$  electrons
3. Four  $\pi$  electrons
4. Unpaired electrons in P orbitals

Question Number : 61 Question Id : 4646621361 Question Type : MCQ

Arrange the following molecules in order of increasing bond length :



Options :

1.  $\text{O}_2 < \text{O}_2^- < \text{O}_2^+ < \text{O}_2$
2.  $\text{O}_2^+ < \text{O}_2^- < \text{O}_2 < \text{O}_2^{2-}$
3.  $\text{O}_2 < \text{O}_2^+ < \text{O}_2^- < \text{O}_2^{2-}$
4.  $\text{O}_2^+ < \text{O}_2 < \text{O}_2^- < \text{O}_2^{2-}$

Question Number : 62 Question Id : 4646621362 Question Type : MCQ

Particles requiring antisymmetric wave functions are called as \_\_\_\_\_.

Options :

1. Bosons
2. Fermions
3.  $\alpha$ -particles
4. Photons

Question Number : 63 Question Id : 4646621363 Question Type : MCQ

The Pauli antisymmetry principle states that the total wave function for electrons must be :

Options :

1. Antisymmetrical
2. Symmetrical
3. Constant
4. Zero

Question Number : 64 Question Id : 4646621364 Question Type : MCQ

In Perturbation theory, the first order correction to energy depends on :

Options :

1. Square perturbation operator
2. Unperturbed wave function and perturbation operator
3. Only the Kinetic energy operator
4. Spin angular momentum only

Question Number : 65 Question Id : 4646621365 Question Type : MCQ

Which description best fits for Pz orbital ?

Options :

1. A single sphere with no nodes
2. Two lobes along z-axis with a nodal plane through the nucleus
3. Four lobes in a plane
4. A ring around the nucleus

Question Number : 66 Question Id : 4646621366 Question Type : MCQ

For a quantum mechanical Harmonic oscillator :

Options :

1. The energy levels are not equally spaced
2. Ground state energy is zero
3. The energy levels are separated by  $n h \nu$  where  $n > 1$
4. The ground state energy is not equal to zero

Question Number : 67 Question Id : 4646621367 Question Type : MCQ

The correct IUPAC name of

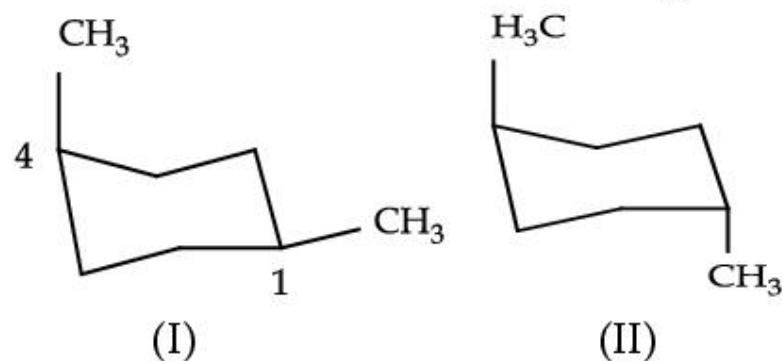
$\text{CH}_2 = \text{C}(\text{CH}_3)\text{CH} = \text{CH}_2$  is :

Options :

1. 2-Methylbuta-1, 3-diene
2. Isoprene
3. 3-Methylbuta-1, 3-diene
4. 2-Methylbutadiene

Question Number : 68 Question Id : 4646621368 Question Type : MCQ

Which statement is correct for the given 1, 4-dimethyl cyclohexane stereoisomers ?

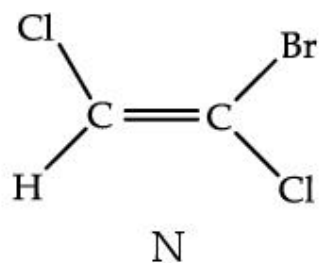
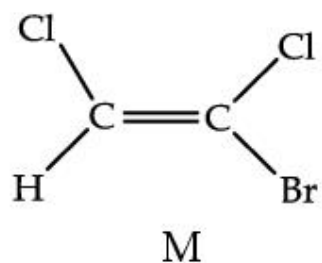


Options :

1. I and II both are trans 1, 4-dimethyl cyclohexane
2. I is cis-1, 4-dimethyl cyclohexane and II is trans-1, 4-dimethyl cyclohexane
3. I is trans-1, 4-dimethyl cyclohexane and II is cis-1, 4-dimethyl cyclohexane
4. I and II both are cis 1, 4-dimethyl cyclohexane

Question Number : 69 Question Id : 4646621369 Question Type : MCQ

The correct notation of double bond of following molecules (M and N) respectively is :



Options :

1. E and E
2. E and Z
3. Z and Z
4. Z and E



Question Number : 70 Question Id : 4646621370 Question Type : MCQ

Which statement regarding cyclohexane conformation is **correct** ?

Options :

1. Boat conformation is more stable than chair due to lower torsional strain
2. Chair conformation is free from angle strain and torsional strain
3. Twist boat is less stable than boat
4. Ring flipping converts cis-1, 2-dimethyl cyclohexane into trans-1, 2-dimethyl cyclohexane

Question Number : 71 Question Id : 4646621371 Question Type : MCQ

Which conformation of Methyl groups in 2-chloro 1, 3-dimethyl cyclohexane will be most stable ?

Options :

1. 1-axial, 3-equatorial
2. 1-axial, 3-axial
3. 1-equatorial, 3-equatorial
4. 1-equatorial, 3-axial

Question Number : 72 Question Id : 4646621372 Question Type : MCQ

Which statement is **false** about geometrical isomers ?

Options :

1. Geometrical isomers have same connectivity but differ in spatial arrangement
2. Cis isomers are always more stable than trans isomers
3. Trans-2-butene is more stable than cis-2-butene
4. Geometrical isomers may differ in dipole moment and boiling point

Question Number : 73 Question Id : 4646621373 Question Type : MCQ

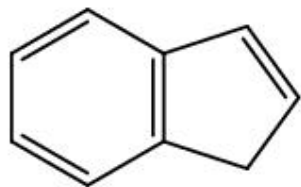
Which statement regarding azulene is **correct** ?

Options :

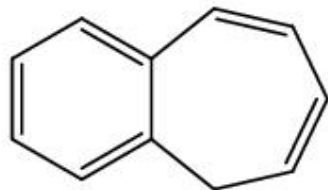
1. It is benzenoid aromatic compound
2. It contains  $8\pi$  electrons and is antiaromatic
3. It is a non-benzenoid aromatic hydrocarbon possessing  $10\pi$  electrons
4. It is non-aromatic because it lacks a benzene ring

Question Number : 74 Question Id : 4646621374 Question Type : MCQ

The loss of proton in compound (A) and (B) :



(A)



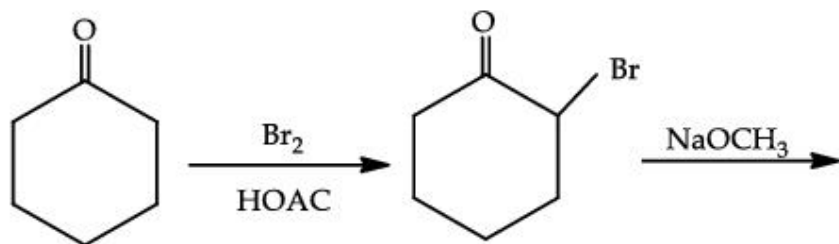
(B)

Options :

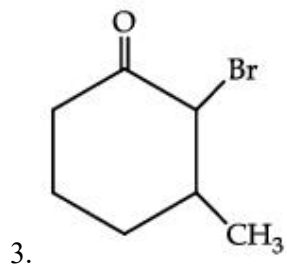
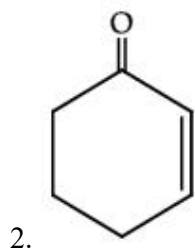
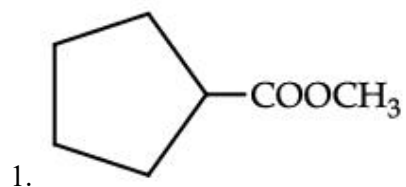
1. (A) gives aromatic anion while (B) gives antiaromatic anion
2. (A) gives antiaromatic anion while (B) gives aromatic anion
3. (A) and (B) both gives aromatic anion
4. (A) and (B) both gives antiaromatic anion

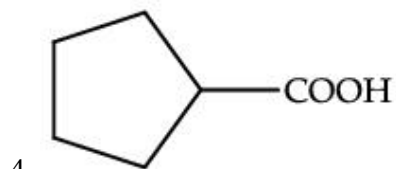
Question Number : 75 Question Id : 4646621375 Question Type : MCQ

The major product formed in the following reaction is :



Options :





Question Number : 76 Question Id : 4646621376 Question Type : MCQ

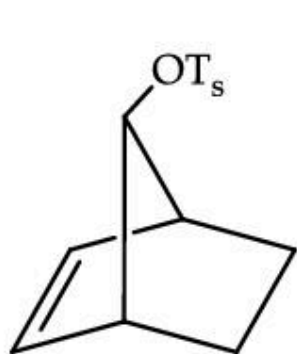
Which of the following **correctly** describes a triplet carbene ?

Options :

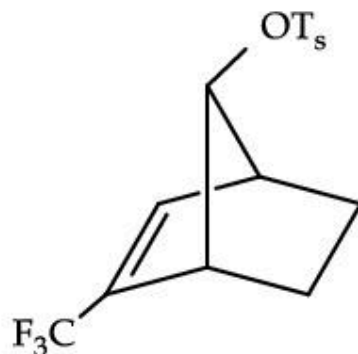
1. Two paired electrons in an ' $sp^2$ ' orbital
2. One electron in ' $sp^2$ ' orbital and one in a ' $p$ ' orbital
3. None electron in the ' $p$ ' orbital
4. Four valence electrons around carbon

Question Number : 77 Question Id : 4646621377 Question Type : MCQ

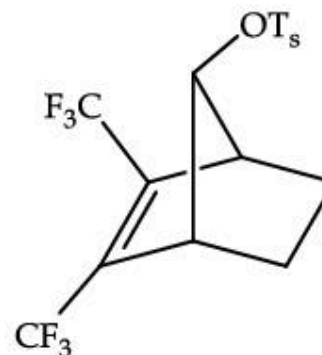
The correct order for the rate of acetolysis among the following molecules is :



A



B



C

Options :

1.  $B > C > A$
2.  $A > B > C$
3.  $C > B > A$
4.  $A > C > B$

Question Number : 78 Question Id : 4646621378 Question Type : MCQ

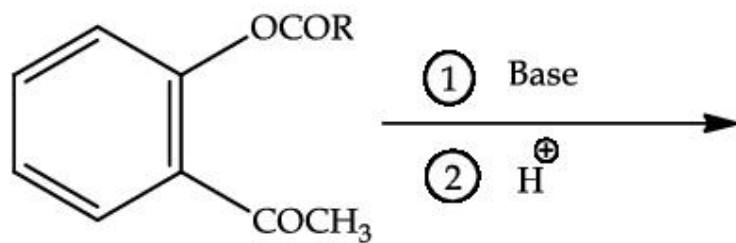
When (R)-2-bromobutane is treated with sodium methoxide in methanol at room temperature, the major product is formed through E2 pathway. Which one of the following statement is correct ?

Options :

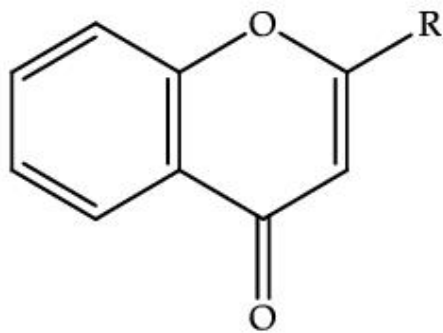
1. The major alkene formed is 1-butene because Hofmann elimination predominates.
2. The major alkene formed is trans-2-butene due to anti-periplanar elimination.
3. The reaction proceeds through a carbocation intermediate, giving racemized products.
4. The reaction gives cis-2-butene exclusively due to syn-elimination.

Question Number : 79 Question Id : 4646621379 Question Type : MCQ

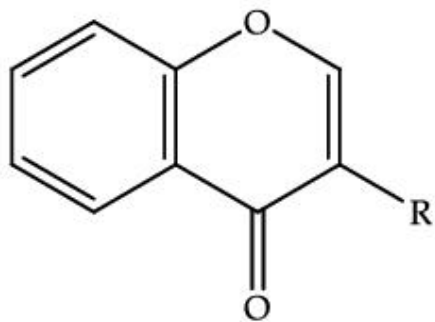
The major product formed in the following reaction is :



Options :

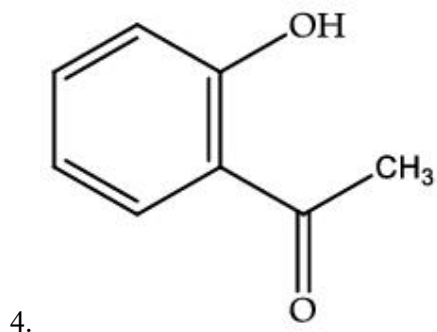
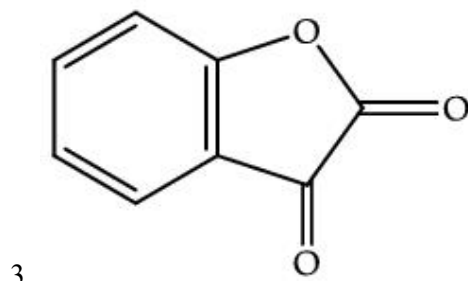


1.



2.





Question Number : 80 Question Id : 4646621380 Question Type : MCQ

Which of the following represents rate equation of a typical E2 elimination reaction ?

Options :

1.  $\text{Rate} = K [\text{alkyl halide}]$
2.  $\text{Rate} = K [\text{alkyl halide}]^2$
3.  $\text{Rate} = K [\text{alkyl halide}] [\text{base}]$
4.  $\text{Rate} = K [\text{alkyl halide}]^2 [\text{base}]$

Question Number : 81 Question Id : 4646621381 Question Type : MCQ

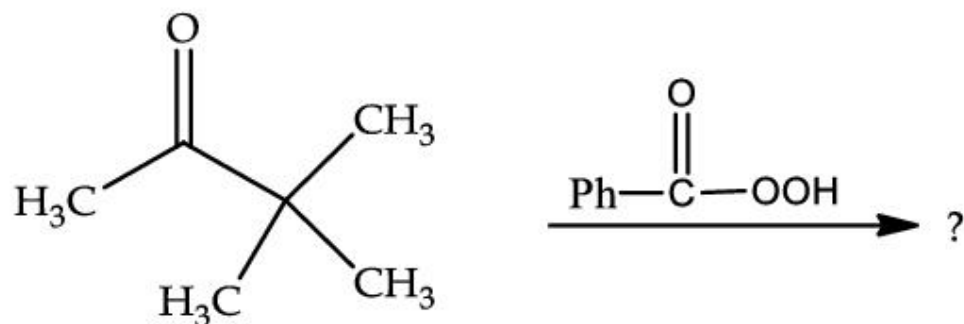
An optically active Ketoxime undergoes Beckmann rearrangement under acidic condition. Which statement regarding the migrating group is correct ?

Options :

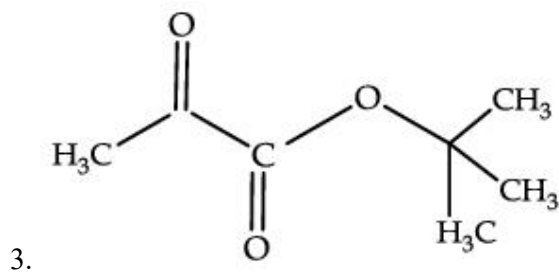
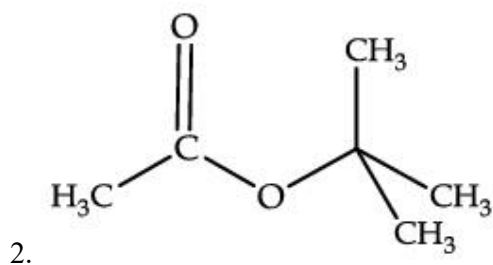
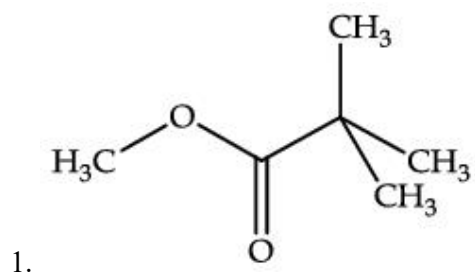
1. The group with greater electron donating ability always migrates
2. The group anti(trans) to the leaving hydroxyl group migrates
3. The smaller alkyl group preferentially migrates
4. Migration occurs randomly, leading to racemization

Question Number : 82 Question Id : 4646621382 Question Type : MCQ

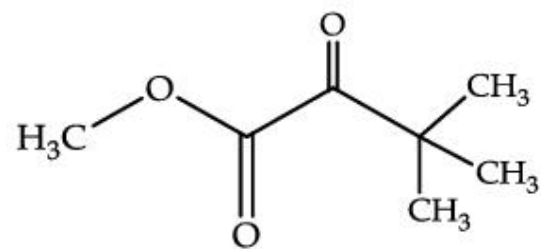
The product obtained in the reaction of methyl t-butyl ketone with perbenzoic acid is :



Options :

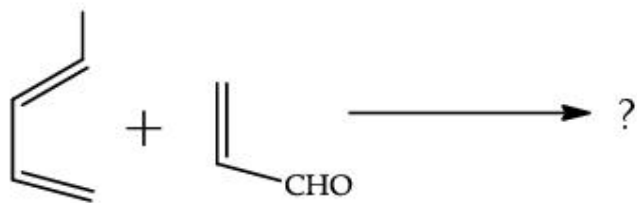


4.

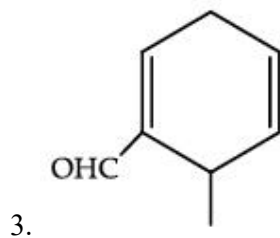
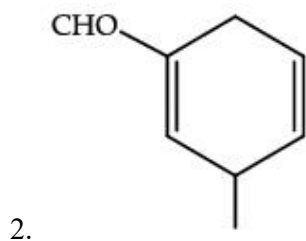
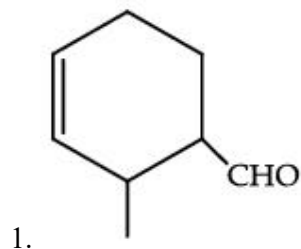


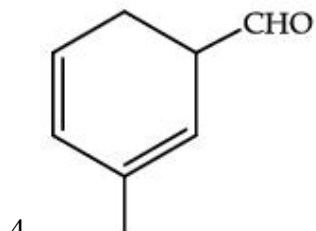
Question Number : 83 Question Id : 4646621383 Question Type : MCQ

What is the major product in the following reaction ?



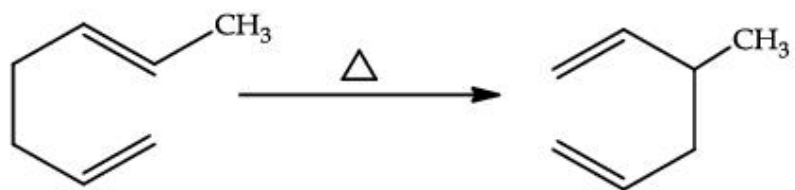
Options :





Question Number : 84 Question Id : 4646621384 Question Type : MCQ

The following reaction is an example of :

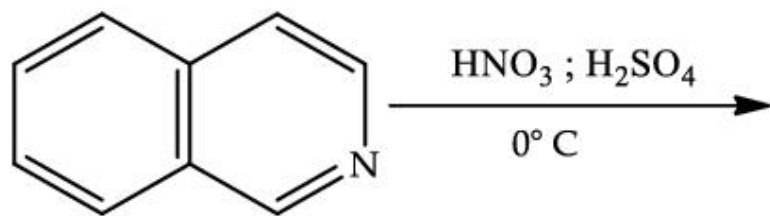


Options :

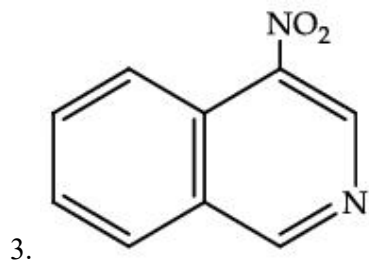
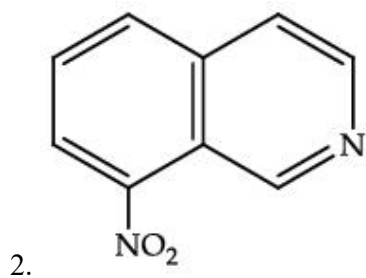
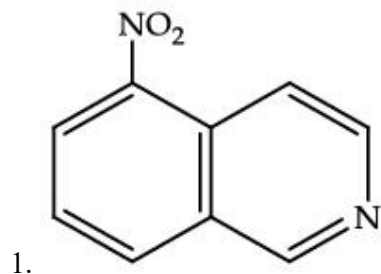
1. Electrocyclic Rearrangement
2. [1, 5] – Sigmatropic Rearrangement
3. [1, 3] – Sigmatropic Rearrangement
4. [3, 3] – Sigmatropic Rearrangement

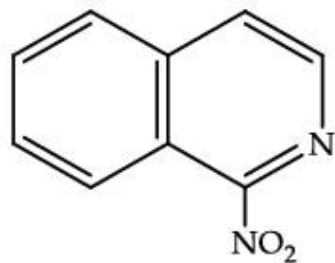
Question Number : 85 Question Id : 4646621385 Question Type : MCQ

The major product formed in a following reaction is :



Options :





4.

Question Number : 86 Question Id : 4646621386 Question Type : MCQ

Nucleophilic substitution in pyridine occurs preferentially at which position ?

Options :

1. C-4 only
2. C-3 only
3. C-2 only
4. All positions equally



Question Number : 87 Question Id : 4646621387 Question Type : MCQ

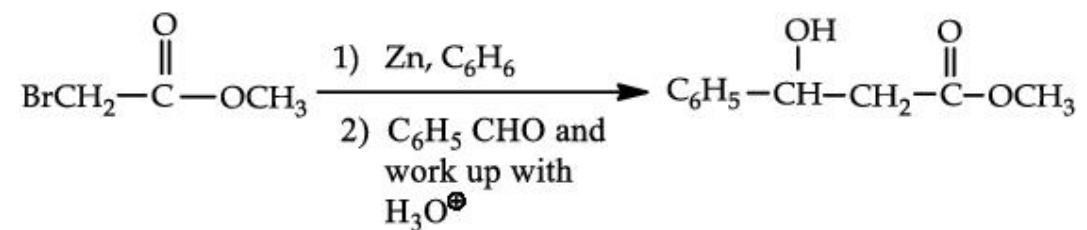
Which of the following is an important methods for the synthesis of Quinoline consists of heating a mixture of aniline, glycerol, conc.  $\text{H}_2\text{SO}_4$  and ferrous sulphate in the presence of an oxidizing agent ?

Options :

1. Gaureschi Synthesis
2. Fischer Synthesis
3. Gabriel Synthesis
4. Skraup's Synthesis

Question Number : 88 Question Id : 4646621388 Question Type : MCQ

The name of following reaction is :

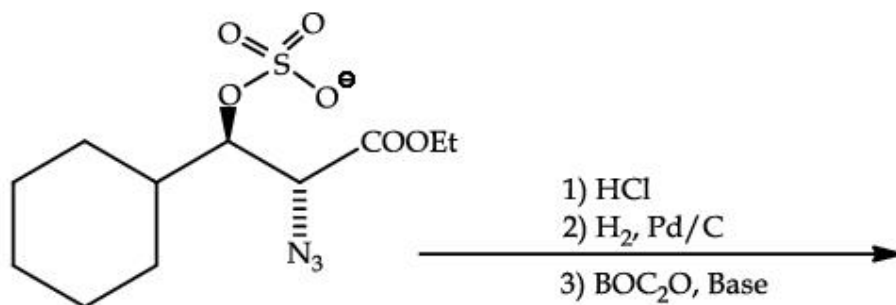


Options :

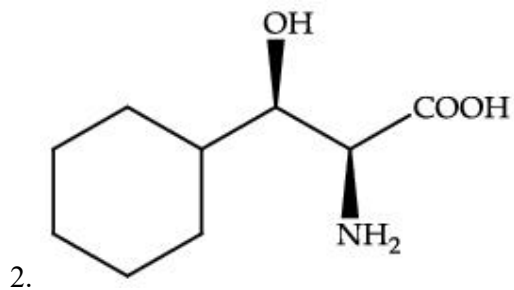
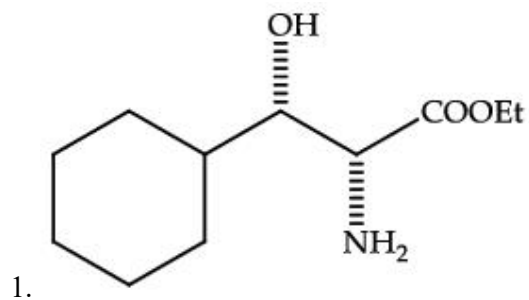
1. Wittig Reaction
2. Aldol Reaction
3. Reformatsky Reaction
4. Grignard Reaction

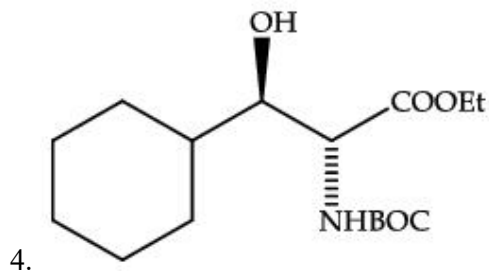
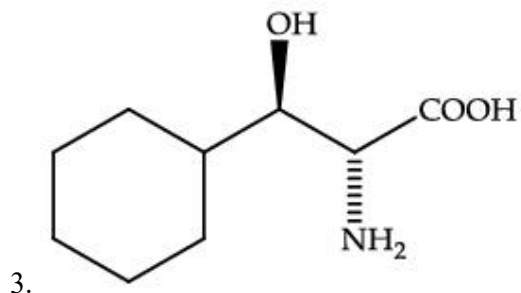
Question Number : 89 Question Id : 4646621389 Question Type : MCQ

Major product form in the following reaction is :



Options :





Question Number : 90 Question Id : 4646621390 Question Type : MCQ

The idealized fragments resulting from disconnection of target molecules which needs to be replaced by reagent is termed as :

Options :

1. Intermediates
2. Synthons
3. Functional groups
4. Substrate

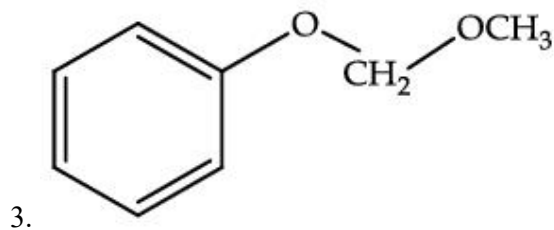
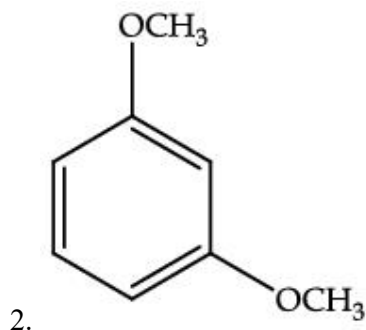
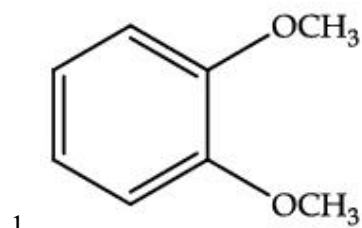
Question Number : 91 Question Id : 4646621391 Question Type : MCQ

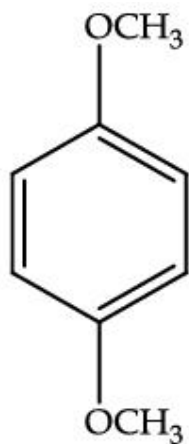
A compound with molecular formula  $C_8H_{10}O_2$  shows following  $^1H$  NMR spectra :

6.80 (4H, Singlet) ; 3.75 (6H, Singlet).

The correct structure is :

Options :





4.

Question Number : 92 Question Id : 4646621392 Question Type : MCQ

Which is the peak observed in the mass spectrum of 2-octanone, due to McLafferty rearrangement?

Options :

1. 58 m/z

2. 43 m/z

3. 128 m/z

4. 113 m/z

Question Number : 93 Question Id : 4646621393 Question Type : MCQ

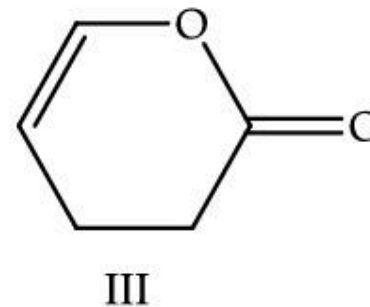
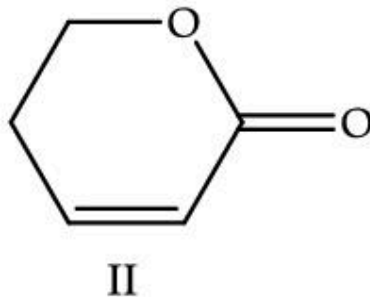
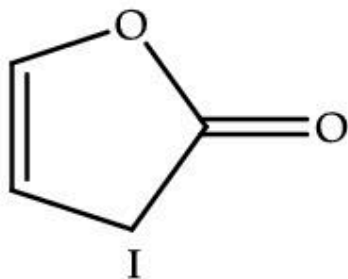
In the  $^1\text{H}$  NMR spectrum of ethyl acetate, the methyl group attached directly to the carbonyl carbon appears as :

Options :

1. Quartet near  $\delta$  4.1 ppm
2. Triplet near  $\delta$  1.2 ppm
3. Singlet near  $\delta$  2.0 ppm
4. Doublet near  $\delta$  2.5 ppm

Question Number : 94 Question Id : 4646621394 Question Type : MCQ

The correct order for the stretching frequency of carbonyl group of the following molecules is :



Options :

1.  $II > I > III$
2.  $III > I > II$
3.  $I > III > II$
4.  $I > II > III$



Question Number : 95 Question Id : 4646621395 Question Type : MCQ

The  $\beta$ -glycoside bond in nucleotides connects :

Options :

1. The base and phosphate group
2. The sugar and phosphate group
3. The sugar and nitrogenous base
4. Two phosphate groups

Question Number : 96 Question Id : 4646621396 Question Type : MCQ

Which statement is **correct** regarding transition states in asymmetric synthesis ?

Options :

1. Lower-energy transition states lead to major products
2. Higher-energy transition states are always preferred
3. All transition states have the same energy
4. Transition-state energy does not affect product distribution

Question Number : 97 Question Id : 4646621397 Question Type : MCQ

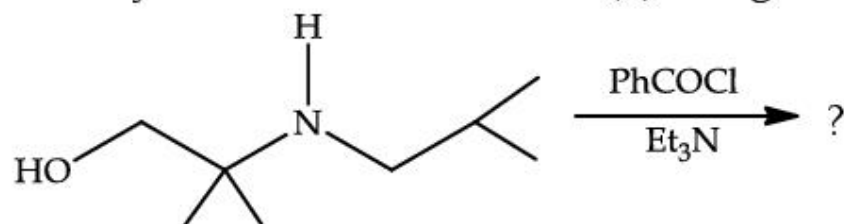
Which reagent is used in Sharpless epoxidation ?

Options :

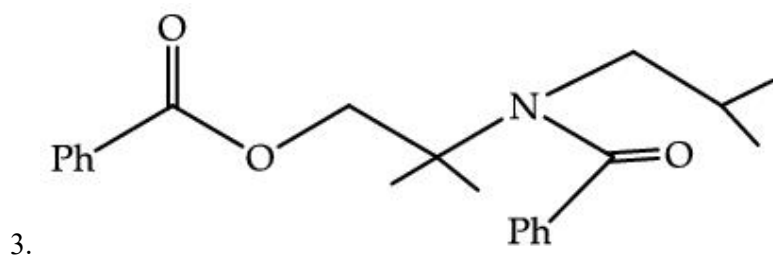
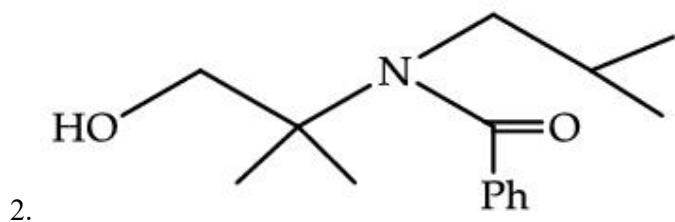
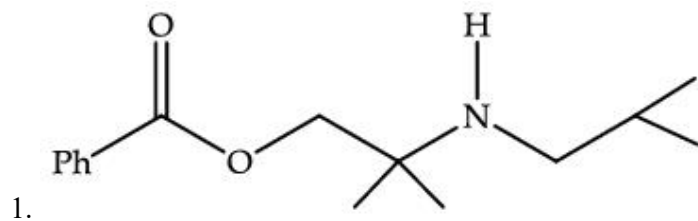
1.  $\text{Ti}(\text{O}^i\text{Pr})_4$  and tartrate ester
2.  $\text{KMnO}_4$
3.  $\text{BF}_3 \cdot \text{Et}_2\text{O}$  and  $\text{NaOH}$
4. m-CPBA

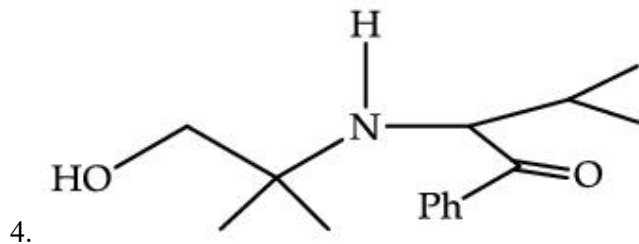
Question Number : 98 Question Id : 4646621398 Question Type : MCQ

The acylation of amino alcohol (x) using benzoyl chloride in presence of triethylamine gives :



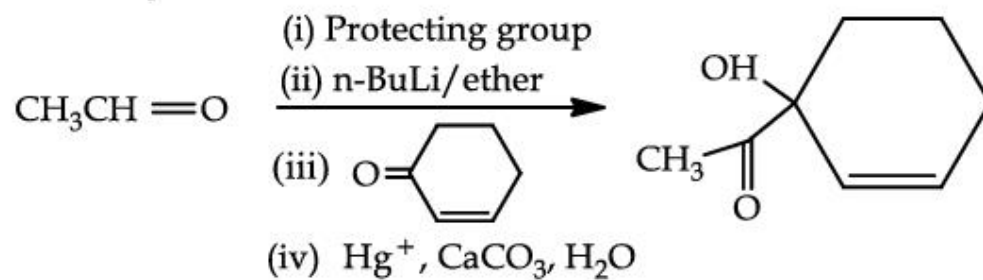
Options :





Question Number : 99 Question Id : 4646621399 Question Type : MCQ

In the synthesis of 'B' molecule....



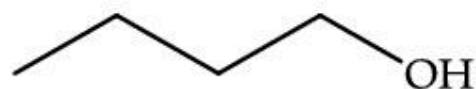
the protecting group is used as :

Options :

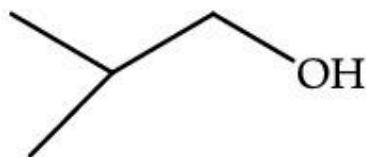
1. Propane diol
2. Ethylene glycol
3. Propane dithiol
4. Ethylene dithiol

Question Number : 100 Question Id : 4646621400 Question Type : MCQ

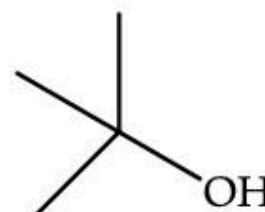
The three alcohols given below has mol. formula  $C_4H_{10}O$ .



(A)



(B)



(C)

Chemical shifts in  $C^{13}$  NMR peaks of methyl groups are 15.2, 20.4 and 32.7 ( $\delta$  : ppm). The signal at 15.2 is identified from :

Options :

1. (A) and (B)
2. (B) and (C)
3. Only (A)
4. Only (C)